

Tejas Kadam

Vellore, Tamil Nadu, India

tejaskadam209@email.com — +91-8591877007 — GitHub — LinkedIn

Education

Vellore Institute of Technology, Vellore

2024 – 2028

B.Tech in Computer Science and Business Systems

Relevant Coursework: Data Structures, Algorithms, Machine Learning, Statistics, Database Systems, Linear Algebra, Probability

Technical Skills

Programming Languages: Python, C++, SQL

Machine Learning: Neural Networks, Transformers, Natural Language Processing (NLP), Reinforcement Learning, Time-Series Modeling, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Fine-tuning

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Hugging Face Transformers, Flask, FastAPI

Web Development: React, Node.js, REST APIs, JavaScript, D3.js

Databases: MongoDB, SQL

Tools & Platforms: Git, Linux, Data Pipelines, API Development

Hardware / Embedded Systems: Raspberry Pi, Arduino, ESP32

Projects

Artemis – AI System for Financial Call Compliance Analysis

Code

- Built an AI system to analyze financial call recordings and detect regulatory compliance risks using NLP.
- Implemented audio preprocessing pipeline including speaker diarization, language detection, and speech-to-text transcription.
- Applied transformer-based models for entity extraction and intent detection in financial conversations.
- Designed a compliance scoring module to identify commitments and potential regulatory risks.

ShopTrack – Retail Sales Forecasting and Analytics Platform

Code

- Developed a time-series forecasting system to predict category-wise retail sales trends.
- Designed modular pipeline for preprocessing, forecasting, and prediction filtering.
- Built a web analytics dashboard with interactive visualizations and MongoDB-based storage.

SpendWise – Personal Finance Dashboard

- Developed a full-stack web application for tracking expenses and managing financial categories.
- Built financial visualizations using D3.js to analyze spending patterns.
- Implemented a recommendation engine suggesting optimized spending strategies based on user budgets.

Digit Recognition using Neural Networks from Scratch

Code — Live

- Implemented a neural network from scratch in Python using NumPy to classify handwritten digits (MNIST dataset).
- Developed forward propagation, Leaky ReLU activation, softmax output layer, and manual backpropagation.
- Built and deployed an interactive interface allowing users to draw digits and obtain predictions.

Reinforcement Learning Grid Visualizer

Code — Live

- Implemented a Q-learning agent to learn optimal paths in a customizable grid environment.
- Built visualizations to display state-value functions and learning progression during training.
- Developed an interactive interface allowing users to experiment with RL parameters.

Gesture Recognition using Gyroscope Sensors

- Collected motion data from gyroscope sensors to train a gesture recognition model.
- Integrated the trained model with Raspberry Pi for gesture-based home automation.

Machine Learning Experience

- Fine-tuned transformer models for aspect-based sentiment analysis on text datasets.
- Implemented preprocessing pipelines and evaluation metrics for NLP classification tasks.
- Built neural networks from scratch using NumPy to understand optimization and training dynamics.

Achievements

- Finalist in two university-level hackathons among multiple competing engineering teams.
- Solved 370+ algorithmic problems on LeetCode focusing on data structures and algorithms.
- Secured 99.4 percentile in Physics in JEE Main examination.
- Achieved 98.6 percentile in Class 10 board examinations.

Research and Learning

- Studied neural network implementation from *Make Your Own Neural Network* by Tariq Rashid.
- Exploring reinforcement learning, computational statistics, and time-series modeling.
- Practicing advanced algorithmic problem solving including dynamic programming.